

Deep Generative Inverse Design of Auxetic Metamaterials: Navigating Multi-Objective Demands and Out-of-Distribution Generalization

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Abstract

Deep generative models can propose metamaterial unit cells for a target effective property in milliseconds. A usable design, however, must simultaneously match the target after binarization and finite-element (FE) verification, be manufacturable, and ideally be obtainable for targets outside the training data. We study these three demands for two-dimensional auxetic unit cells generated by a conditional variational autoencoder (cVAE) fine-tuned with differentiable homogenization. For accuracy, we show that test-time latent refinement must optimize the verified design: refining the continuous decoder output drives the objective to $\sim 10^{-10}$ while the verified error stays six orders of magnitude larger, whereas embedding the verification operators in the objective raises the FE-verified R^2 of ν_{12} on 100 held-out targets from 0.874 to 0.998 (single sample) and from 0.986 to 0.997 (best of 30). For multi-objective selection, a weighted accuracy–manufacturability–aesthetics score agrees moderately with an independent Pareto ranking (median Spearman 0.71). For out-of-distribution targets, the cVAE generates designs with the correct, unseen sign of the Poisson ratio where nearest-neighbour retrieval cannot, but it does not extrapolate magnitudes beyond the training range.

Keywords: Auxetic metamaterials; Deep generative models; Differentiable physics; Inverse design; Topology optimization; Homogenization

1 Introduction

Auxetic metamaterials contract laterally under uniaxial compression. Their negative Poisson’s ratio originates in the microarchitecture rather than the chemistry of the constituent material [1–3]. Inverse homogenization by topology optimization is the classical route from a prescribed effective tensor to a unit cell [4–6], but every new target requires an iterative optimization with hundreds of FE solves. Data-driven inverse design amortizes this cost: variational autoencoders, generative adversarial networks and diffusion models trained on simulated libraries return candidate architectures almost instantly [7–14].

Practical use places three demands on such a generator, and they are rarely examined together. (i) *Verified accuracy.* What is manufactured is a binarized, resampled version of the generated density field, so accuracy must be judged by FE analysis of that design, not by a learned surrogate. Surrogates of the forward map are known to be exploitable by optimizers and generators [15]. Single samples also rarely meet tight tolerances, so pipelines generate many candidates and select among them with FE [13]. (ii) *Multiple objectives.* A design that meets the target but contains floating islands, sub-resolution struts or edges that do not tile is unusable. Accuracy therefore has to be traded against manufacturability and, at lower priority, against geometric regularity. (iii) *Out-of-distribution (OOD) targets.* The main argument for a generative model over a lookup in the training library is that it might produce designs for targets the library does not contain. Whether it does, and along which directions of property space, is an empirical question.

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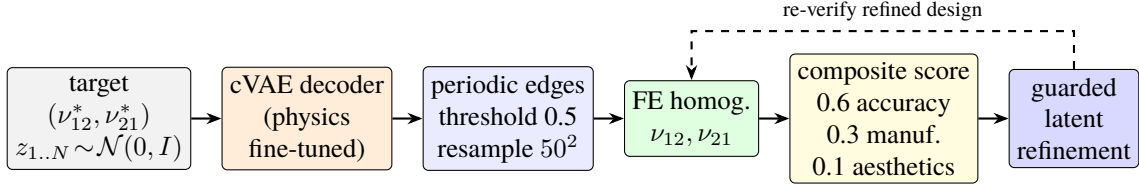


Figure 1: Inverse-design pipeline. Every candidate is verified by FE homogenization after the same post-processing that a manufactured design would undergo. Selection combines verified accuracy with manufacturability (connectivity, minimum feature size, periodicity) and a low-weight aesthetic score. Refinement is detailed in Fig. 2.

This paper addresses the three demands with one pipeline and one evaluation protocol, in which every reported number is FE-verified on the binarized design and every comparison carries a paired confidence interval. Our contributions are:

1. **Binarization-aware latent refinement.** We refine the latent code of a frozen cVAE using exact analytic gradients of a periodic homogenization solver. We show that refinement is only reliable when the objective is evaluated on the verified design, obtained through periodic-edge enforcement, Heaviside projection with continuation [16, 17] and nearest-neighbour resampling. We also provide direct evidence of the grey-density exploitation that occurs otherwise.
2. **Multi-objective selection with an external check.** We rank candidates by a weighted score of accuracy, manufacturability and aesthetics. We test this score against an independent Pareto ranking instead of assuming that it is adequate.
3. **A physically grounded OOD characterization.** We show that symmetric targets with $\nu_{12} = \nu_{21} \leq -1$ are inadmissible for orthotropic cells, which invalidates naive “extreme” benchmarks. We then separate generalization in the *sign* of the Poisson ratio, where the cVAE succeeds and retrieval fails, from generalization in *magnitude*, where all methods saturate at the edge of the training data.
4. **Controlled negative results.** Neither a Kolmogorov–Arnold network (KAN) head nor an implicit wavelet (WIRE) decoder improves on the convolutional baseline under FE-verified metrics.

2 Results

2.1 Framework and evaluation protocol

Figure 1 shows the pipeline. A cVAE maps a target pair (ν_{12}^*, ν_{21}^*) and a latent vector $z \sim \mathcal{N}(0, I)$ to a 64×64 density image. The cVAE is trained in two stages, the second with a loss computed by FE homogenization of prior samples (Methods). For each target, N latents are decoded and post-processed. Post-processing consists of periodic-edge enforcement, thresholding at 0.5 and nearest-neighbour resampling to the 50×50 FE grid. Each candidate is then FE-homogenized and scored for manufacturability and aesthetics. The best candidate by a composite score is optionally refined in latent space, and a refined design is kept only if its verified FE error is lower (“guarded” refinement).

All accuracy numbers are FE-verified on the binarized design. The in-distribution benchmark (IN100) comprises 100 held-out test conditions drawn with a fixed seed. Comparisons report paired bootstrap 95% confidence intervals (CI) over conditions [18]. Success criteria were fixed before each experiment (Methods).

2.2 Verified accuracy requires optimizing the verified design

Continuous refinement exploits grey densities. Refining the continuous decoder output (30 L-BFGS steps [19], gradients through the decoder and the exact homogenization sensitivities) helps single samples of the production model. On IN100, $R^2(\nu_{12})$ rises from 0.874 to 0.976 and the mean absolute error (MAE) of ν_{12} falls by 57.7% (95% CI 49.4–64.6%; Table 1, Fig. 3b). The same refinement, however, *worsens* the

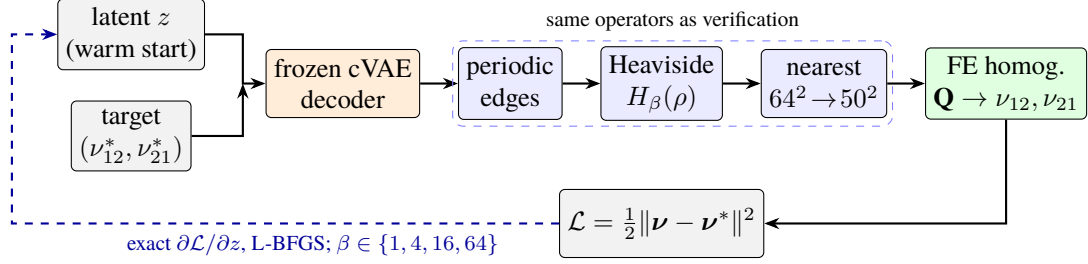


Figure 2: Binarization-aware latent refinement. The frozen decoder output passes through differentiable counterparts of the verification operators before FE homogenization, so the quantity optimized is the quantity verified. The continuous variant omits the dashed block and uses bilinear resampling. Verification replaces H_β by a hard threshold at 0.5.

Table 1: FE-verified accuracy on IN100 (100 held-out targets, production cVAE). ΔMAE is the relative reduction of $\text{MAE}(\nu_{12})$ versus the unrefined design of the same mode, with a paired bootstrap 95% CI (10 000 resamples). Negative values mean that refinement made results worse. FE calls are per condition and include selection.

Mode	Refinement	$R^2(\nu_{12})$	$\text{MAE}(\nu_{12})$	$R^2(\nu_{21})$	ΔMAE [95% CI]	FE calls
Single sample	none	0.874	0.0517	0.477	—	1
	continuous	0.976	0.0219	0.643	+57.7% [49.4, 64.6]	51
	binarization-aware	0.998	0.0043	0.883	+91.8% [89.0, 93.9]	78
Best-of-30	none	0.986	0.0152	0.802	—	30
	continuous	0.968	0.0246	0.637	−62.5% [−98.3, −33.9]	79
	binarization-aware	0.997	0.0057	0.712	+62.6% [50.7, 72.1]	107
	guarded	0.998	0.0047	—	+69.1% [61.6, 75.4]	107

best-of-30 winner. $\text{MAE}(\nu_{12})$ rises from 0.0152 to 0.0246, a change of −62.5% (CI −98.3 to −33.9%), and R^2 drops from 0.986 to 0.968. The objective trace explains why (Fig. 4). At the end of continuous refinement the median objective is 2.5×10^{-10} , but the median verified squared error of the binarized design is 4.4×10^{-4} . The optimizer matches the target on a field whose intermediate densities do not survive thresholding. This is the same failure mode as surrogate exploitation, arising inside an exact physics model.

Binarization-aware refinement. Placing differentiable counterparts of the verification operators inside the objective (Fig. 2; 32 steps over $\beta \in \{1, 4, 16, 64\}$) reduces the objective–verification gap by three orders of magnitude (Fig. 4). All pre-registered in-distribution criteria are met (Table 1).

- **Single samples.** $R^2(\nu_{12})$ rises from 0.874 to 0.998 and $\text{MAE}(\nu_{12})$ falls by 91.8% (CI 89.0–93.9%). The mean relative error of ν_{12} drops from 14.4% to 2.6%, and $R^2(\nu_{21})$ rises from 0.477 to 0.883.
- **After best-of-30.** $\text{MAE}(\nu_{12})$ falls by 62.6% (CI 50.7–72.1%) and $R^2(\nu_{12})$ rises from 0.986 to 0.997. Guarded refinement accepts 92% of refinements and reduces $\text{MAE}(\nu_{12})$ by 69.1% (CI 61.6–75.4%), without extra FE solves.

After best-of-30, $\text{MAE}(\nu_{21})$ decreases from 0.0327 to 0.0128, whereas $R^2(\nu_{21})$ falls from 0.802 to 0.712. A few conditions therefore acquire larger ν_{21} deviations, which the guarded rule, based on the summed error, does not always reject. Refinement costs 77 FE solves per condition on average. With 50–100 ms per solve including sensitivities on the 50×50 grid, this amounts to a few seconds per design.

2.3 Multi-objective demands: manufacturability and composite ranking

Accurate designs are not automatically buildable. Each candidate is checked for (a) connectivity without struts thinner than two pixels and (b) periodicity, meaning that opposite edges must match so that the

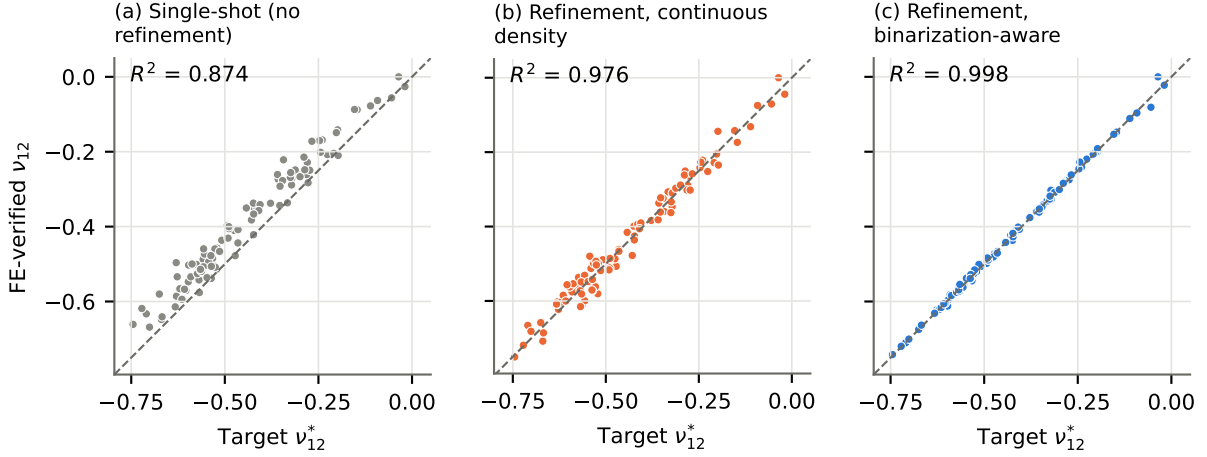


Figure 3: FE-verified versus target ν_{12} on IN100 ($n = 100$, single samples of the production cVAE): (a) unrefined; (b) after continuous refinement; (c) after binarization-aware refinement. All panels start from the same latent code per condition. Dashed line: identity.

Table 2: Multi-objective indicators. Upper block: production pipeline with periodic enforcement and best-of-30 selection on 300 test targets. Manufacturable fraction: share of generated samples passing connectivity, minimum-feature and periodicity checks. Hit rate: fraction of targets whose first sample is auxetic ($\nu_{12} < 0$). Lower block: agreement of the composite score with Pareto ranks (24 targets $\times$ 30 candidates).

Model	$R^2(\nu_{12})$, best-of-30	Single-sample hit rate	Manufacturable fraction
Production cVAE (dataset v2)	0.995	0.997	0.247
Earlier cVAE (dataset v1)	0.998	0.983	0.280
Composite score vs. Pareto rank: Spearman mean 0.693, median 0.714, range 0.43–0.89; 24/24 targets positive.			

cell tiles by translation (Methods). On an earlier cVAE without periodic enforcement, only 0–3.5% of generated samples passed both checks. This rate was uniform across property space, so the problem lies in decoder behaviour rather than in data coverage. Enforcing periodicity by averaging opposite edges before binarization is a training-free operation that raised the pass rate by more than an order of magnitude. It is part of every verification in this paper. With it, 24.7% of the samples of the production model pass all checks (300 test targets, $N = 30$; Table 2). In-distribution, periodic enforcement therefore does not force a trade-off against accuracy.

Among the candidates for a target, selection maximizes the composite score $0.6 s_{\text{acc}} + 0.3 s_{\text{man}} + 0.1 s_{\text{aes}}$. The weights encode the priority order accuracy $>$ manufacturability $>$ aesthetics; they are a design choice, not a learned quantity. To test whether this scalarization respects the multi-objective structure, we computed, for 24 targets with 30 candidates each, the Spearman correlation between the composite score and the non-dominated sorting rank over the three raw objectives (Fig. 5). All 24 correlations were positive (range 0.43–0.89). The median was 0.714, but the mean of 0.693 fell just short of the pre-registered threshold of 0.7. The composite score therefore agrees moderately, not strongly, with the Pareto structure. That the selected candidate always lies on the first Pareto front is guaranteed mathematically for positive weights, so it is not evidence.

2.4 Out-of-distribution generalization

Admissible targets. Before testing extrapolation we asked which targets exist. For a two-dimensional orthotropic material, positive definiteness of the compliance requires $\nu_{12}\nu_{21} < 1$ [20]. Symmetric targets $\nu_{12}^* = \nu_{21}^* \leq -1$ are therefore infeasible. In the training data the most negative $\nu_{12} = -1.95$ is paired with $\nu_{21} = -0.04$, and the largest product $\nu_{12}\nu_{21}$ is 0.43. This matters for benchmarking. An earlier “extreme

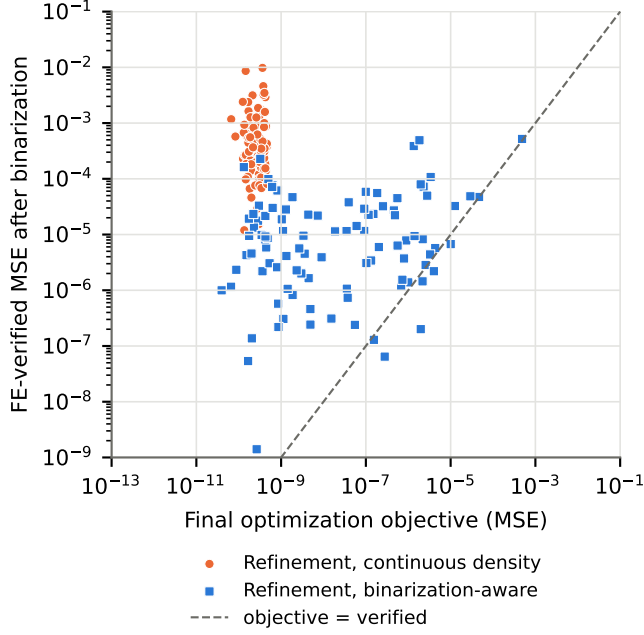


Figure 4: Objective–verification gap. Each point is one IN100 condition. The horizontal axis shows the final refinement objective; the vertical axis shows the squared Poisson-ratio error of the binarized design measured by an independent FE solve. Continuous refinement (circles) reaches objectives near 10^{-10} while verified errors remain at 10^{-4} – 10^{-2} . Binarization-aware refinement (squares) moves points towards the identity. Values below 10^{-14} are clipped for display.

Table 3: Sign-flip OOD targets (training data contain only $\nu_{12} < 0$). FE-verified ν_{12} of nearest-neighbour retrieval and of the cVAE (first sample and best of 10); last column: manufacturable fraction of the 10 cVAE samples.

(ν_{12}^*, ν_{21}^*)	Retrieval	cVAE, first	cVAE, best of 10	Manuf.
(0.1, 0.1)	−0.002	0.082	0.104	0.6
(0.3, 0.3)	−0.002	0.183	0.204	0.6
(0.5, 0.5)	−0.002	0.191	0.280	0.8
(0.8, 0.8)	−0.002	0.241	0.301	0.6
(0.2, 0.5)	−0.002	0.225	0.194	0.9
(0.5, 0.2)	−0.002	0.109	0.177	0.6
MAE(ν_{12})	0.402	0.237	0.191	

auxetic” test in our project used targets such as $(-2.1, -2.1)$ and $(-2.2, -1.6)$, all six of which violate the bound. Its failures therefore say nothing about extrapolation, and we exclude them here. Out-of-range magnitude targets below are anisotropic, with $\nu_{21}^* = -0.05$.

Sign: the cVAE generalizes, retrieval cannot. Every training sample is auxetic ($-1.95 \leq \nu_{12} \leq -0.002$). For six non-auxetic targets with $0.1 \leq \nu_{12}^* \leq 0.8$, nearest-neighbour retrieval from the 57 216-sample training library necessarily returns the sample closest to zero ($\nu_{12} = -0.002$), with the wrong sign for every target. The cVAE (best of 10) produced positive ν_{12} for all six targets (Table 3, Fig. 6), with MAE 0.19 versus 0.40 for retrieval. The attained magnitude, however, saturates near 0.3. For $\nu_{12}^* = 0.8$ the cVAE reached 0.30. The cVAE thus extrapolates the *direction* of the property but not its full magnitude. This test used an earlier checkpoint (dataset v1) and six targets, and should be read as qualitative evidence.

Magnitude: all methods saturate at the training boundary. We swept $\nu_{12}^* \in \{-1.0, -1.5, -1.75, -2.0, -2.25, -2.5\}$ ($\nu_{21}^* = -0.05$) with ten independent latent draws per target, best-of-30 selection and periodic enforcement (Table 4, Fig. 7). Inside the training range, binarization-aware refinement lowers the

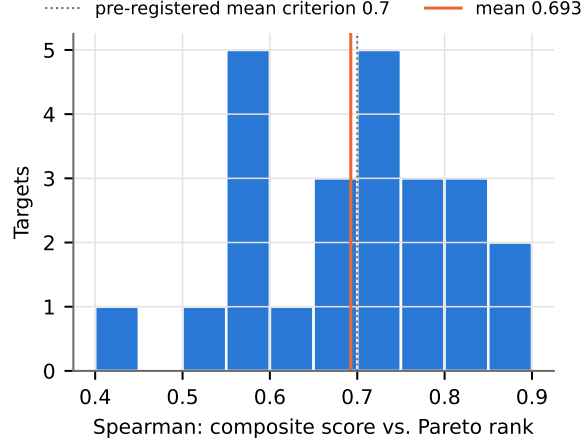


Figure 5: Distribution over 24 targets of the Spearman correlation between the composite score and the Pareto rank of 30 candidates (objectives: verified $|\Delta\nu_{12}|$, manufacturability score, aesthetic score). Dotted: pre-registered threshold for the mean.

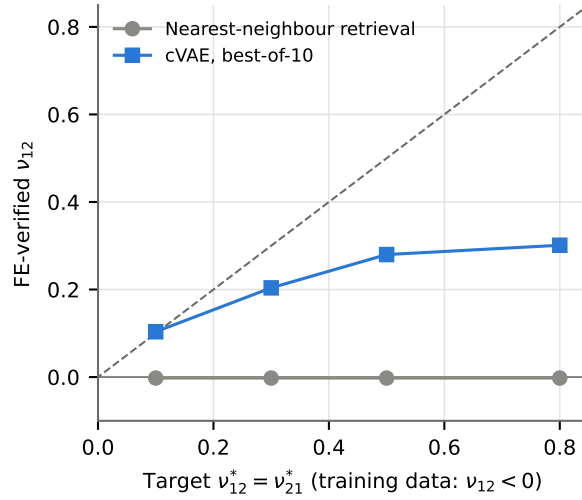


Figure 6: Sign-flip generalization for symmetric non-auxetic targets. Retrieval is bounded by the training library ($\nu_{12} \leq -0.002$). The cVAE reaches the correct sign but saturates in magnitude. Dashed: identity.

median relative error of ν_{12} to 0.2–0.4%. Beyond the training minimum of -1.95 , every variant saturates. The most negative verified ν_{12} among all 30 designs at $\nu_{12}^* = -2.5$ was -1.99 . Unguarded refinement is then worse than no refinement (32.4% versus 22.5% at -2.5). The guarded variant meets the pre-registered 8% criterion at -2.0 (4.9%), but only because this target lies 0.05 beyond the boundary; from -2.25 onwards it merely matches best-of-30. Refinement is a local search in the decoder’s latent space and cannot create magnitudes the decoder never learned to represent.

2.5 Architecture ablations

KAN versus linear heads. We replaced the three linear layers of the cVAE (latent mean, log-variance and decoder bottleneck) with Kolmogorov–Arnold layers [21]. Both heads were trained with identical recipes, hyperparameters and seeds (123 and 7). KAN was declared the winner only if the paired CI excluded zero for both seeds. It did not win in any mode (Table 5). In the production mode (best-of-30 without refinement), the linear head was significantly better for both seeds. With refinement, both heads reached $R^2(\nu_{12}) = 0.997\text{--}0.999$. The KAN head has $3.2\times$ more parameters (4.79 M versus 1.49 M) and needs about $1.6\times$ the time per epoch, so we retain the linear head.

Table 4: Anisotropic magnitude sweep ($\nu_{21}^* = -0.05$; best-of-30 with periodic enforcement; 10 repeats per target). Entries: median relative error of FE-verified ν_{12} . The training range of ν_{12} ends at -1.95 .

ν_{12}^*	-1.0	-1.5	-1.75	-2.0	-2.25	-2.5
Best-of-30, no refinement	0.5%	0.9%	4.1%	11.0%	16.1%	22.5%
+ continuous refinement	1.4%	1.3%	3.2%	36.5%	38.9%	43.9%
+ binarization-aware refinement	0.4%	0.2%	0.3%	12.5%	22.3%	32.4%
+ binarization-aware, guarded	0.2%	0.2%	0.3%	4.9%	14.8%	21.9%

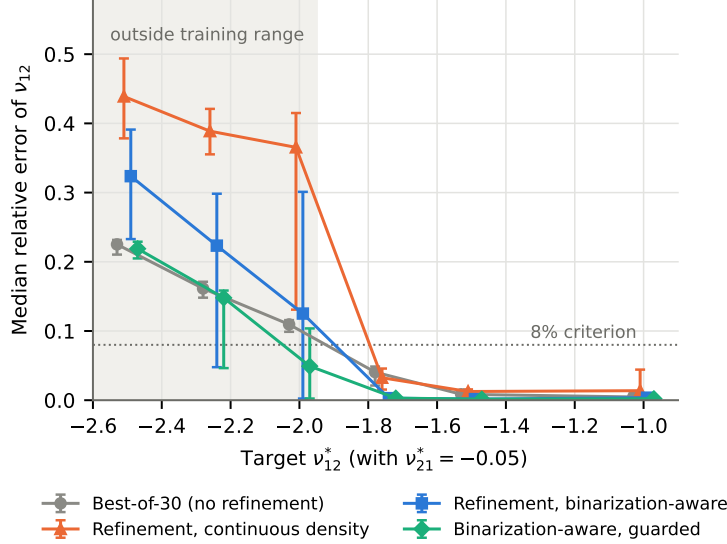


Figure 7: Median relative error of FE-verified ν_{12} versus target ($\nu_{21}^* = -0.05$). Error bars: interquartile range over 10 latent draws per target; markers offset horizontally for legibility. Shaded: outside the training range. Dotted: pre-registered 8% criterion at $\nu_{12}^* = -2.0$.

Implicit wavelet decoder. A resolution-agnostic WIRE decoder [22] was given one final attempt. It combined Heaviside projection inside the physics loss, no forced symmetry and the two-stage recipe. The pre-registered criteria were a single-sample hit rate of at least 30% and a positive FE R^2 on 24 held-out targets. It reached a hit rate of 12.5% and an FE R^2 of -7.19 after best-of-30, against about 99.7% hit rate for the convolutional decoder. Only 4.6% of its samples were manufacturable. We therefore closed this direction.

3 Discussion

Accuracy. The central methodological finding is that physics-based refinement of a generated design must evaluate its objective on the same object that will be verified. Differentiating an exact solver removes surrogate error but not the mismatch between a continuous decoder output and a binarized unit cell. Fig. 4 shows that this mismatch lets the optimizer reach objectives of 10^{-10} on a field that the verifier sees very differently. The problem is invisible in the loss curve and only surfaces when a strong baseline is refined. The ingredients of the remedy are standard in density-based topology optimization [16, 17]. What is new is applying them to the output of a frozen generative model during latent search, together with the exact resampling and periodic post-processing used at verification. The same principle should extend to any post-processing step that is included in the verification.

Multiple objectives. The pipeline handles the three objectives at different stages. Periodicity is enforced by construction, connectivity and feature size enter through the composite score, and accuracy is improved

Table 5: Controlled KAN versus linear ablation on IN100 (two seeds). ΔMAE : relative reduction of $\text{MAE}(\nu_{12})$ of KAN versus linear (positive favours KAN), paired bootstrap 95% CI. Bold: CI excludes zero.

Mode	$R^2(\nu_{12})$ Linear / KAN		ΔMAE KAN vs. Linear	
	seed 123	seed 7	seed 123	seed 7
Single sample	0.852 / 0.904	0.815 / 0.911	+13.3% [−0.2, 24.4]	+ 18.2% [3.8, 30.4]
Single sample + refinement	0.994 / 0.997	0.998 / 0.999	+33.2% [−15.1, 60.5]	+ 33.9% [11.7, 52.3]
Best-of-30	0.977 / 0.974	0.982 / 0.980	− 25.2% [−56.2, −1.0]	− 19.9% [−42.1, −1.7]
Best-of-30 + refinement	0.998 / 0.997	0.998 / 0.999	+11.3% [−38.7, 43.6]	+ 44.1% [24.9, 59.7]

last by refinement that keeps the periodic edges. This ordering is pragmatic, and our evidence for it is partial. The composite score agrees only moderately with the Pareto ranking. The rank-based normalization of the accuracy term within each candidate pool is a plausible cause that we have not yet tested. We also have not measured whether refinement changes connectivity or feature size. Designers who need guaranteed manufacturability can still restrict selection to candidates that pass all checks. In earlier experiments this required large candidate pools ($N \approx 1500$) and lowered accuracy.

Out-of-distribution generalization. Our results suggest a more precise version of the common claim that generative models “generalize”. Along the sign of the Poisson ratio, the cVAE produces designs in a region with zero training density, which a lookup method cannot do by construction. Along magnitude, the generator, its refinement and best-of- N selection all stop at the training boundary. OOD claims should therefore state the direction in property space and should first check that the target is physically admissible. Magnitude extrapolation will likely require new training data near the boundary, for example from topology optimization seeded by generated designs, rather than better sampling from the current model.

Limitations. (i) All analyses use linear-elastic, small-strain homogenization of two-dimensional cells on a 50×50 grid with a base-material Poisson ratio of 0.3. Large strain and three-dimensional architectures are not addressed. (ii) Designs are verified numerically, including against an independent FE implementation, but not experimentally. (iii) The sign-flip test used six targets and an earlier checkpoint. The magnitude sweep used ten repeats per target. (iv) The aesthetic score is a simple heuristic (mirror symmetry and boundary smoothness) and has not been validated against human judgement. (v) In-distribution, nearest-neighbour retrieval is highly competitive in accuracy, so the advantage of a generative model shown here is specific to out-of-distribution sign changes. (vi) Refinement adds about 77 FE solves per target. This is inexpensive in 2D but would require efficient adjoint solvers in 3D.

4 Methods

4.1 Data generation

Unit cells were generated by SIMP topology optimization [23–25] with a density filter [26], periodic boundary conditions and energy-based homogenization [27–29]. Auxetic objectives were minimized from four seed topologies, using the optimality-criteria update or the method of moving asymptotes [30] depending on the seed. Process parameters (volume fraction, penalization, filter radius, move limit, void size, lattice rotation) were screened by Latin hypercube sampling [31] and refined in an adaptive multi-batch design of experiments using Sobol’ sequences [32] and optimized Latin hypercubes. Samples with oscillatory convergence, degenerate topologies or labels outside the physical range were removed. The 50×50 density fields were box-filtered to 64×64 images and labelled with the homogenized (ν_{12}, ν_{21}) . The production dataset contains 13 624 unique cells, split 70/15/15 by seed family (2 044 test samples). Training images are augmented sixfold by rotations and reflections, with $\nu_{12} \leftrightarrow \nu_{21}$ swapped under 90° rotations, giving 57 216 training samples.

4.2 Homogenization and exact sensitivities

Density fields $\rho \in [0, 1]^{50 \times 50}$ are discretized with bilinear quadrilateral elements and the modified SIMP interpolation $E(\rho) = E_{\min} + \rho^p(E_0 - E_{\min})$ with $E_{\min} = 10^{-9}E_0$, $p = 3$ and $\nu_0 = 0.3$. The homogenized stiffness $\mathbf{Q}$ follows from three periodic load cases through element mutual energies [29]. Poisson ratios are obtained from $\mathbf{S} = \mathbf{Q}^{-1}$ as $\nu_{12} = -S_{12}/S_{11}$ and $\nu_{21} = -S_{12}/S_{22}$, which remains valid for rotated, fully anisotropic cells. Because the energy formulation is self-adjoint, the sensitivities $\partial \mathbf{Q} / \partial \rho_e$ are available without extra solves. They are propagated analytically through $\partial \mathbf{S} = -\mathbf{S}(\partial \mathbf{Q})\mathbf{S}$ and the quotient rule in a custom PyTorch [33] autograd function, whose gradients agree with finite differences to a relative error below 10^{-4} . The solver agrees with an independent scikit-fem [34] implementation to 10^{-9} on 24 designs.

4.3 Conditional VAE

The cVAE [35, 36] is conditioned on $c = (\nu_{12}, \nu_{21})$ and has a 32-dimensional latent space. The encoder consists of four convolution–batch-norm–ReLU–pooling blocks (32 to 256 channels) that keep a 4×4 spatial map, followed by linear layers for the mean and log-variance. The decoder mirrors the encoder with transposed convolutions and a sigmoid output. Stage 1 trains with binary cross-entropy reconstruction, a KL term with linear warm-up [37] and a property-consistency term from a frozen convolutional surrogate (weight $\gamma = 20$; surrogate test R^2 of 0.974 for ν_{12} and 0.964 for ν_{21}). Stage 2 resumes from stage 1 and adds the squared error between c and the FE-homogenized Poisson ratios of decoded *prior* samples $z \sim \mathcal{N}(0, I)$ (weight 20; 8 samples every second epoch). This is the regime used at inference, and training with this loss from scratch fails. Optimization uses Adam [38]. Checkpoints are selected by FE-verified R^2 on validation conditions rather than by validation loss, which the KL warm-up makes unreliable.

4.4 Verification, manufacturability and composite score

A decoded image is verified by (1) periodic-edge enforcement, in which opposite boundary columns and then rows are replaced by their mean; (2) thresholding at 0.5; (3) nearest-neighbour resampling to 50×50 with the index map of the imaging library used to build the dataset; and (4) FE homogenization. Manufacturability combines three checks on the binary image: a single connected solid phase, a minimum feature size of two pixels (morphological erosion), and periodicity (at most 10% mismatched pixels between opposite edges). The graded score s_{man} is the mean of the three indicators. The aesthetic score s_{aes} averages mirror symmetry about both axes and an inverse normalized perimeter-to-area ratio computed with periodic wrap-around. The accuracy score s_{acc} is $|\Delta \nu_{12}|$ min–max normalized within the candidate pool, with sign reversed so that higher is better. The Pareto validation applies iterated non-dominated sorting to the three raw objectives and correlates the resulting rank with the composite score per target.

4.5 Latent refinement

Given ν^* and a starting latent z_0 (single sample or best-of- N winner), we minimize

$$\mathcal{L}_\beta(z) = \frac{1}{2} \left\| \nu(\mathcal{R} \circ H_\beta \circ \mathcal{P} \circ D(z, c)) - \nu^* \right\|_2^2, \quad (1)$$

where D is the frozen decoder, $\mathcal{P}$ the (linear) periodic-edge enforcement, $\mathcal{R}$ the nearest-neighbour resampling implemented as a differentiable gather with an identical index map, and

$$H_\beta(\rho) = \frac{\tanh(\beta\eta) + \tanh(\beta(\rho - \eta))}{\tanh(\beta\eta) + \tanh(\beta(1 - \eta))}, \quad \eta = 0.5, \quad (2)$$

the smoothed Heaviside projection [17]. Thirty-two L-BFGS steps (step size 0.5, strong-Wolfe line search) are split evenly across $\beta \in \{1, 4, 16, 64\}$, and L-BFGS is restarted at each level because curvature information from a softer projection does not carry over. The continuous baseline applies the loss to $D(z, c)$ with bilinear resampling for 30 steps. Guarded refinement keeps the refined design only if its verified $|\Delta \nu_{12}| + |\Delta \nu_{21}|$ is smaller than that of the start.

4.6 Out-of-distribution protocols and retrieval baseline

Retrieval returns the training sample with the smallest Euclidean distance to the target in (ν_{12}, ν_{21}) . The sign-flip test used six non-auxetic targets, 10 cVAE samples per target and FE-oracle selection. The magnitude sweep used six anisotropic targets with $\nu_{21}^* = -0.05$ and ten independent repeats of best-of-30 per target. Admissibility was checked against $\nu_{12}\nu_{21} < 1$ [20].

4.7 Statistics and pre-registration

IN100 consists of 100 test conditions drawn with seed 123. Relative MAE reductions and 95% CIs are computed by paired percentile bootstrap over conditions with 10 000 resamples [18]. An improvement is claimed only if the CI excludes zero. Criteria were recorded before each experiment. They were: at least 20% MAE reduction for single samples; no significant MAE increase after best-of-30; a median relative error of at most 8% at $\nu_{12}^* = -2.0$; a mean Spearman correlation of at least 0.7 for the composite score; for KAN, a CI excluding zero for both seeds; and for WIRE, a hit rate of at least 30% with FE $R^2 > 0$. L-BFGS on GPU is not bit-wise deterministic, and repeated runs differed by about 0.006 in R^2 .

Data and code availability

Code, trained checkpoints and all per-condition outputs underlying the tables and figures will be made available at [repository URL to be inserted].

Acknowledgements and competing interests

[To be completed.]

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